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Entity Framing and Role Portrayal in the News

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Motivation

- Most media analysis tools only detect:
 - Who is mentioned
 - Whether the coverage is positive or negative
- They do not capture:
 - How actors are **framed** (e.g., protagonist, antagonist, innocent)
- Focuses on two topics:
 - Climate change
 - Russia—Ukraine conflict

Introduction

- Entity framing captures how news media portray entities as **Protagonists**, **Antagonists**, or **Innocents**, drawing on the Social Construction of Target Populations to shape public perception.
- These roles **activate emotional responses** (e.g., freedom fighter vs. terrorist), influencing how audiences interpret events, especially during conflict or political unrest.

Contributions

- ✓ Release of a novel **multilingual** dataset with **entity-framing** annotations.
- ✓ A comprehensive taxonomy of **22 archetypes** under three main roles.
- ✓ Detailed corpus **statistics and analysis** across languages, domains, and roles.
- ✓ Benchmarks using **fine-tuned** and **zero-shot prompting** strategies with LLMs.

Our Dataset

- Languages: Bulgarian, English, Hindi, European Portuguese, Russian
- Domains: **Ukraine-Russia War, Climate Change**
- Annotations: **5,800+** entity mentions labeled with roles
- Novel Taxonomy: **22 fine-grained roles** under three main categories

Entity Framing: Annotated Example

Putin says what Russia needs to do to win special operation in Ukraine

Russia will win the special operation in Ukraine if the society shows consolidation and composure to the enemy, President Vladimir Putin said during a visit to the Ulan-Ude Aviation Plant on March 14, Rossiya 24 TV channel said.

Underdog

Russia is not improving its geopolitical position in Ukraine. Instead, **Russia** is fighting "for the survival of Russian statehood, for the future development of the country and our children."

"In order to bring peace and stability closer, we, of course, need to show the consolidation and composure of our society. When the enemy sees that our society is strong, internally braced up, consolidated, then, without any doubt we will come to reach what we are striving for — both success and victory," Putin said.

According to him, many of the current problems began after the collapse of the Soviet Union, when they tried to put pressure on

Victim

Russia to "destabilise the internal political situation." "Hordes of international terrorists" new sent to the purpose to accomplish this goal, Putin said.

Afterwards, the West decided to start rehabilitating Nazism in Russia's neighbouring states, including in Ukraine.

Nevertheless, Putin continued, Russia had long tried to build partnerships with both Western countries and Ukraine. However, after 2014, when the West contributed to the coup in Ukraine, the state of affairs changed dramatically. It was then when they started exterminating those who advocated the development of normal relations with Russia, he said.

Guardian

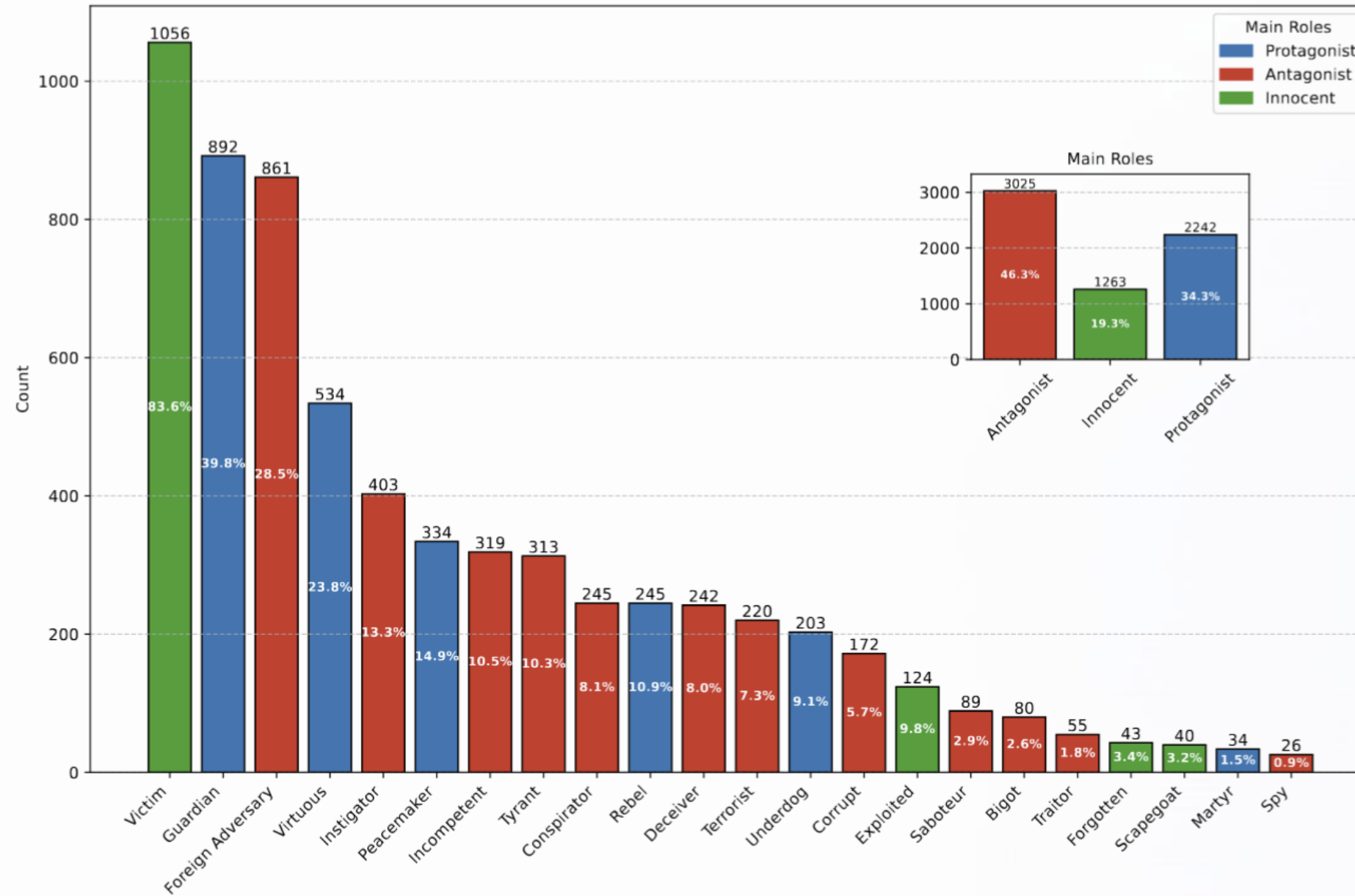
According to Putin, **Russia** was forced to launch the special operation to protect the population. **Western countries** were

Saboteur

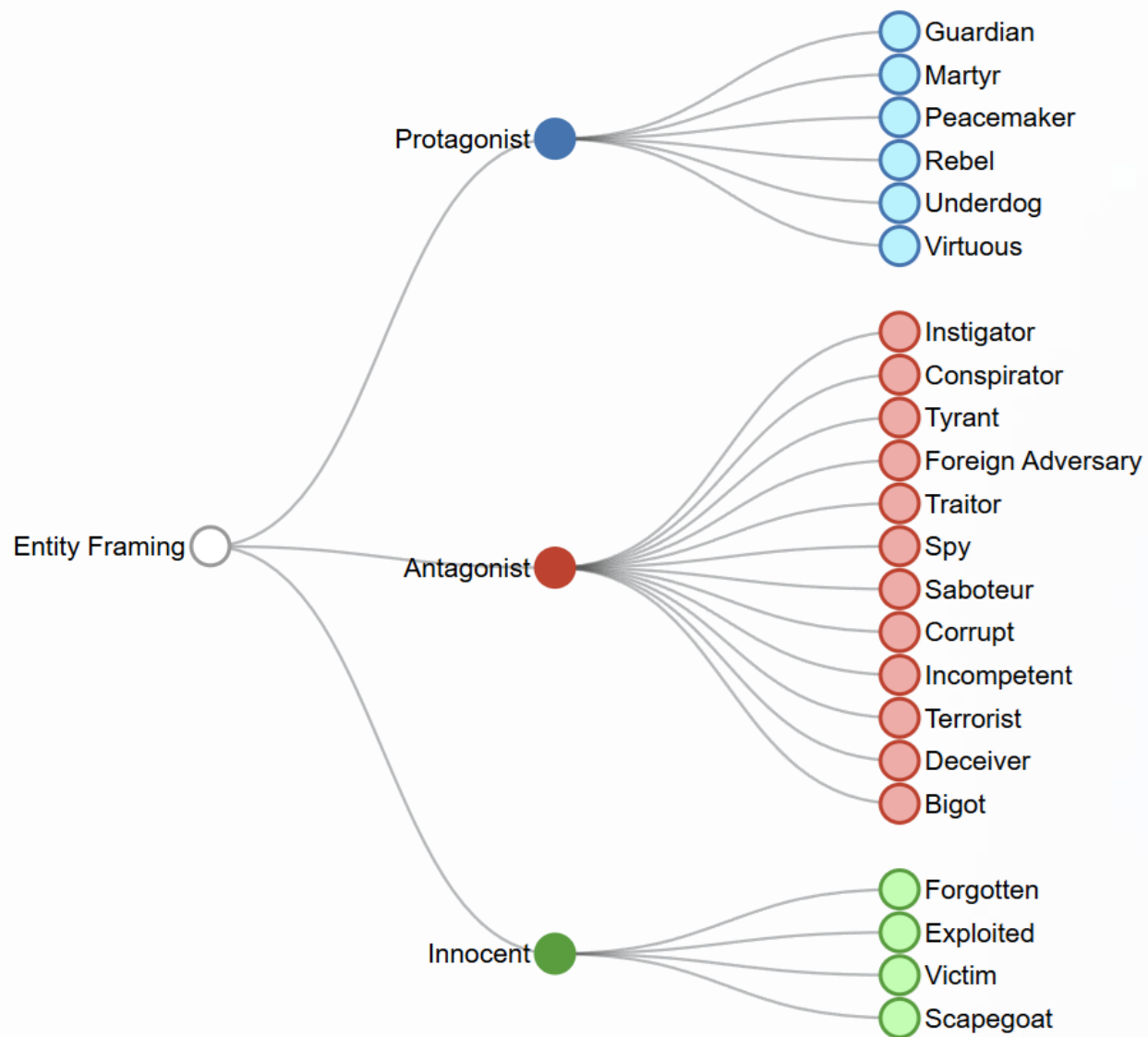
Virtuous

hoping to break Russia quickly, but they were wrong, he said adding that **Russia** managed to raise its economic sovereignty since 2022.

Distribution of Annotations



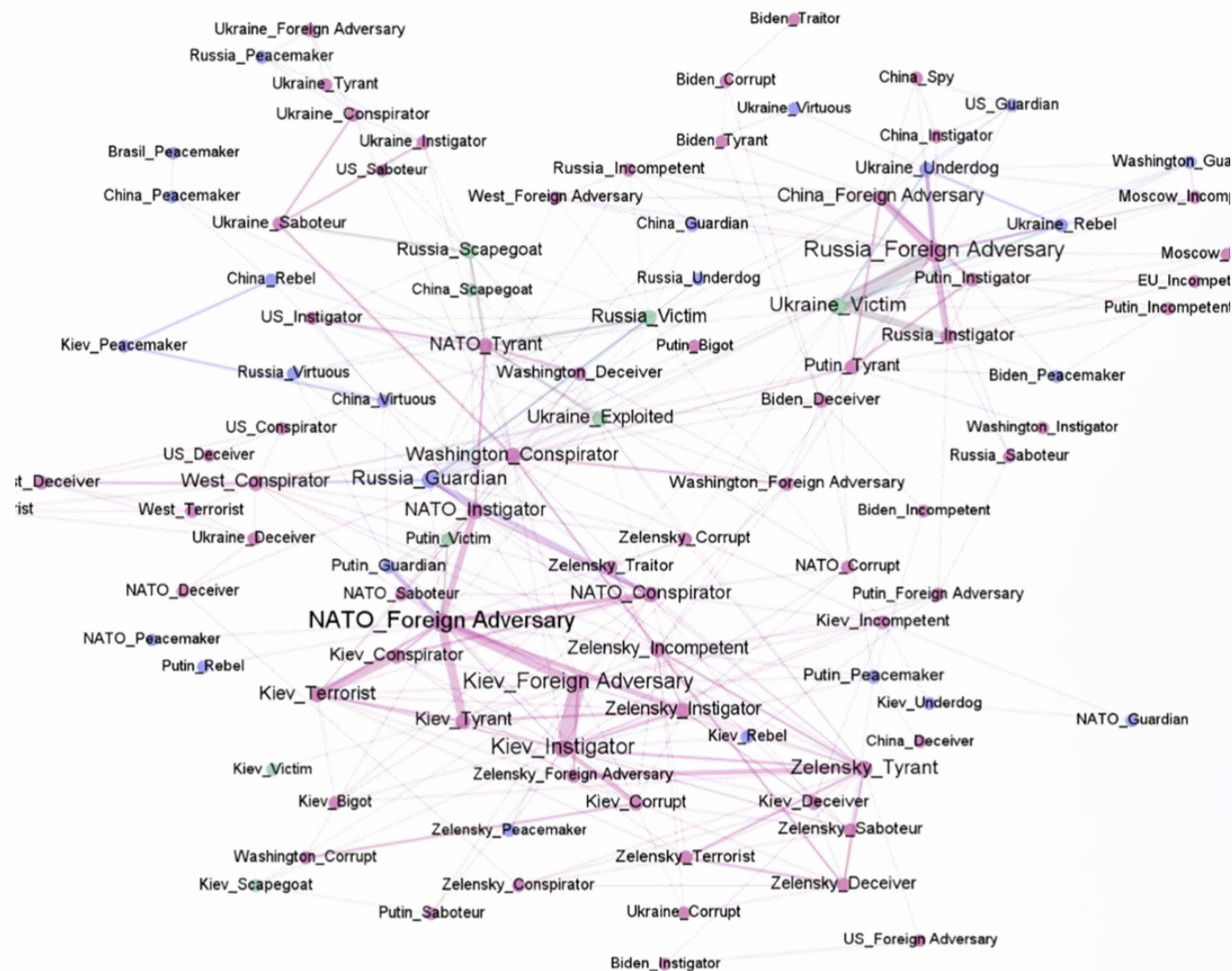
Entity Framing: A Novel Taxonomy



Benchmarks

Method	Lang.	Main Role		Fine Grained Role				Cost (USD)
		Accuracy	Balanced Accuracy	P	R	Micro F1	Macro F1	
Single-Step LLM Prompting	EN	.8346	.6756	.2692	.4632	.3405	.2171	0.7989
	BG	.8065	.7380	.3725	.5588	.4471	.3481	0.2751
	HI	.6327	.6247	.2753	.4000	.3262	.2196	2.4696
	PT	.7812	.7455	.5167	.6643	.5813	.2891	1.0200
	RU	.7558	.6719	.3939	.5843	.4706	.4644	0.7587
	All	.7030	.6957	<u>.3211</u>	<u>.4726</u>	<u>.3824</u>	.3103	5.3223
Multi-Step LLM Prompting	EN	.8031	.6799	.2887	.4118	.3394	.2383	0.5130
	BG	.8031	.6799	.4318	.5588	.4872	.3601	0.5130
	HI	.6367	.6284	.2676	.2868	.2769	.1771	1.4581
	PT	.8125	.7882	.3895	.2643	.3149	.2498	0.5634
	RU	.7442	.6680	.4118	.4719	.4398	.3774	0.4769
	All	<u>.7053</u>	<u>.7017</u>	.3051	.3294	.3168	.2765	3.1852
XLM-R	EN	0.6889	0.5276	.1854	.2828	.2240	.1327	—
	BG	0.7333	0.5791	.3030	.3030	.3030	.1349	—
	HI	0.7025	0.7046	.3234	.4951	.3912	.2043	—
	PT	0.8957	0.8840	.6259	.7480	.6815	.2040	—
	RU	0.8000	0.7604	.4831	.4886	.4859	.2364	—
	All	0.7529	0.7553	.3649	.4985	.4213	.2392	—

Downstream Applications



Thank you

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